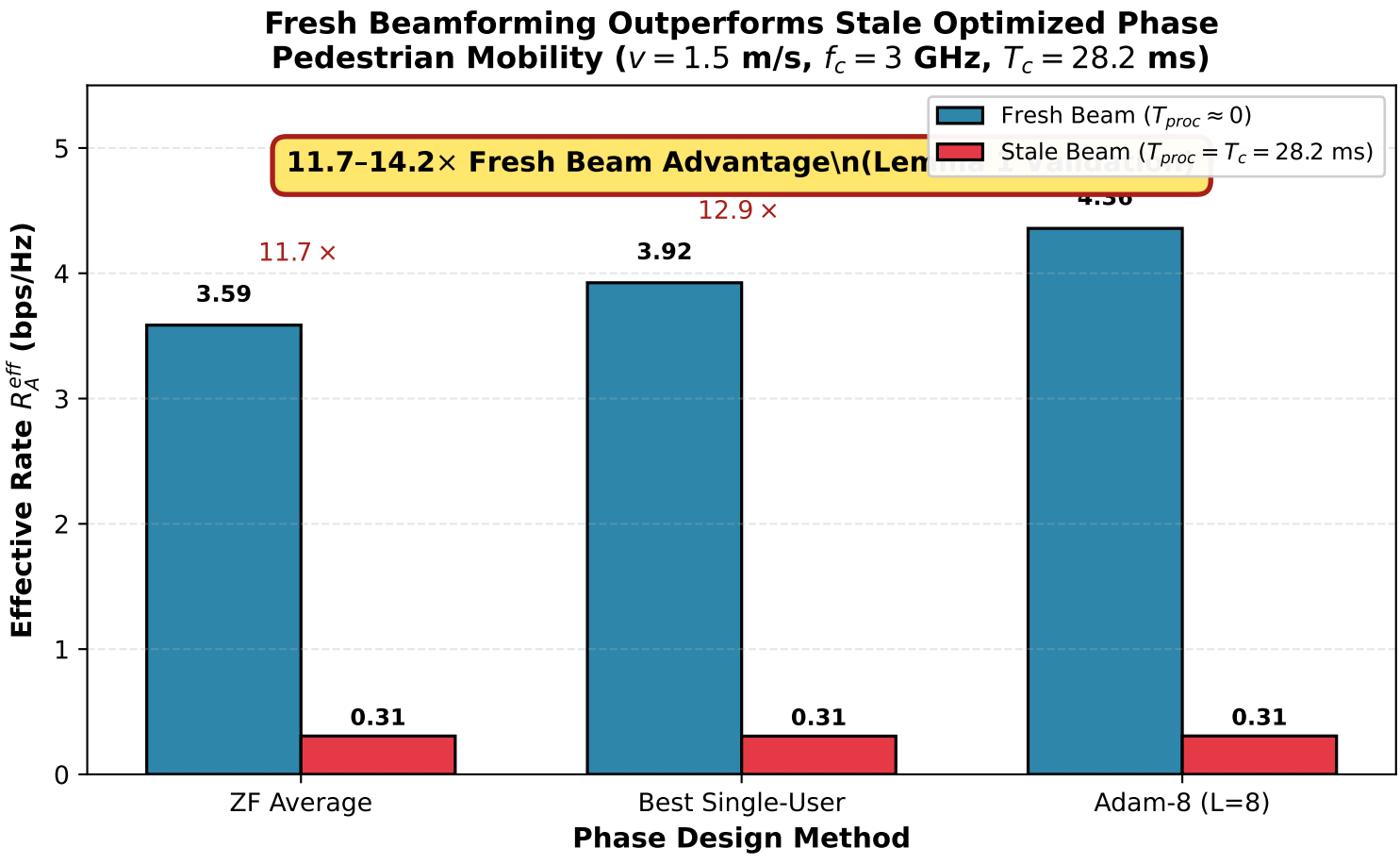
Fresh Beamforming Outperforms Stale Optimized Phase
Pedestrian Mobility ($v = 1.5$ m/s, $f_c = 3$ GHz, $T_c = 28.2$ ms)
Fresh Beam ($T_{proc} \approx 0$)
Stale Beam ($T_{proc} = T_c = 28.2$ ms)
11.7–14.2× Fresh Beam Advantage\n(Len
11.7 ×
12.9 ×
3.59
3.92
0.31
0.31
0.31
Effective Rate R_A^{eff} (bps/Hz)
0
1
2
3
4
5
ZF Average
Best Single-User
Adam-8 (L=8)
Phase Design Method